STRUCTURAL DRAWING SET

BACKYARD CALISTHENICS FORTRESS

Owner-Builder — Town of Bridgewater, Massachusetts

ITEM	VALUE	REFERENCE
Project Location	Bridgewater, MA 02324	—
Zoning Classification	Accessory Structure (< 300 sq ft)	Bridgewater Zoning Ord. Sec. 4.4
Lot Setback Requirement	5 ft rear / side yard minimum	Bridgewater Sec. 4.4.1
Total Footprint	264 sq ft (12x12 Zone A + 10x5 Zone B)	—
Max Height (Zone A)	14 ft	Bridgewater Sec. 4.2.3 (25 ft max)
Max Height (Zone B)	12 ft	Bridgewater Sec. 4.2.3
Frost Depth (per code)	48 inches	780 CMR R403.1.1
Snow Load, Pg	30 psf	Table R301.2(4), IRC 2018
Basic Wind Speed	134 mph, Risk Category II	Table R301.2(4), ASCE 7-16
Seismic Design Category	A (Ss = 0.188g)	ASCE 7-16 Table 1.7-1
Soil Bearing Capacity	2,000 psf (conservative)	—
Live Load — Deck Area	40 psf	IRC Table R301.5

ITEM	VALUE	REFERENCE
Live Load — Play Structure	40 psf, 200 lb dynamic	ASTM F1487 / IRC R301.5
Concrete Strength	3,000 psi, 5% air entrainment	IRC R403.1
Wood Grade	Southern Pine #2 GC PT	AWC NDS 2018
Steel Grade	ASTM A500 Grade B, 46 ksi	AISC 360-16
Adopted Codes	IRC 2018 / IBC 2018 / ASCE 7-16	Mass. 780 CMR 10th Ed.
Permit Required	Yes — Building Permit required	Bridgewater Building Dept
Est. Permit Fee	$48 — $50 (150-300 sq ft tier)	Bridgewater Fee Schedule 7/1/24
Wood Preservative	Copper naphthenate — all field cuts	AWPA M4

SCOPE OF WORK: This drawing set defines the structural system for a backyard calisthenics fortress consisting of two zones: (A) 12 ft x 12 ft adventure tower to 14 ft tall with 13 ft elevated deck and full perimeter safety netting, and (B) 10 ft x 5 ft calisthenics frame to 12 ft tall with pull-up bar, monkey bar ladder, and dip stations. All design loads, member capacities, and connection details comply with IRC 2018, IBC 2018, AWC NDS 2018, and AISC 360-16. For any discrepancy between notes and details, details govern. For discrepancy between plan and elevation, plan governs.

GENERAL NOTES

1.0 GENERAL

– All work shall comply with IRC 2018, IBC 2018, and 780 CMR 10th Edition as adopted by the Town of Bridgewater.
– These drawings are for owner-builder use. A Massachusetts-licensed PE review is recommended for structures exceeding 12 ft in height.
– ALL DIMENSIONS ARE NOMINAL. Actual dimensions are shown on detail drawings.
– DO NOT SCALE DRAWINGS. Use listed dimensions only.
– Coordinate all structural work with underground utilities — call 811 before excavation.

2.0 MATERIALS

– LUMBER: Southern Pine, Structural #2, Grade Mark PT (pressure treated). All cut ends and field modifications to be field-treated with copper naphthenate per AWPA M4.
– CONCRETE: 3,000 psi, 5% air entrainment, slump 4-6 in. Mix per ASTM C94.
– STEEL PIPE: ASTM A500 Grade B, 46 ksi min yield. All steel to be primed with red oxide primer, then finished with exterior latex paint.
– ROPE: 3/4" diameter fibershed climbing rope, min breaking strength 5,000 lb.
– CONNECTORS: Simpson Strong-Tie or equal. All connectors per manufacturer specifications.
– FASTENERS: Hot-dip galvanized (HDG) or stainless steel (304/316). Carbon steel fasteners to be HDG per ASTM A153.
– SAFETY NETTING: 10 ft width, UV-stabilized polyethylene, mesh opening no greater than 2 inches. Installed per ASTM F1487 requirements.
– SHADE SAILS: High-density polyethylene (HDPE), UV-stabilized, minimum 250 gsm, with reinforced grommets at 12 in. spacing.

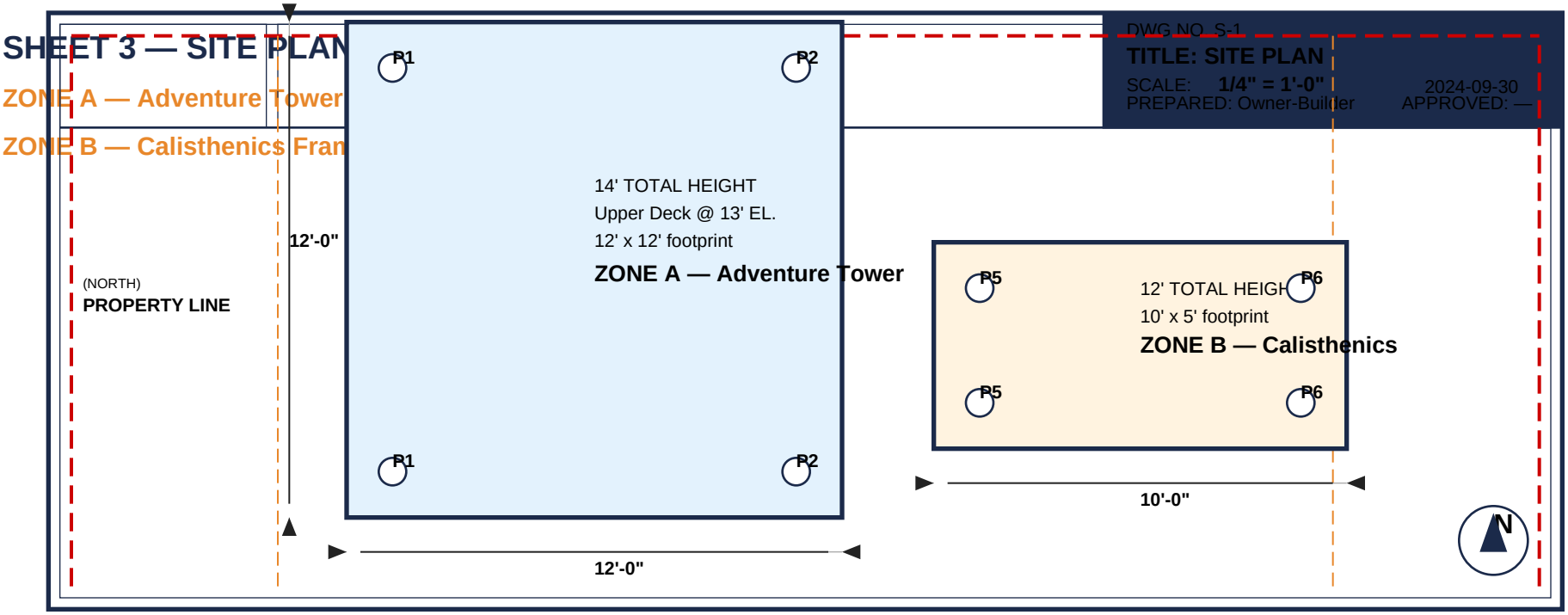
3.0 GENERAL NOTES — STRUCTURAL

– All posts to be set plumb within 1/4" in 10 ft.
– All beams to be leveled within 1/4" in 10 ft span.
– All connections to be verified torqued to manufacturer specifications before inspection.
– Maximum allowable gap between deck boards: 1/4" (per DCA-6 Table 4A).
– All hardware to be stainless steel (304 or 316) or hot-dip galvanized per ASTM A153.

4.0 STANDARD SYMBOLS

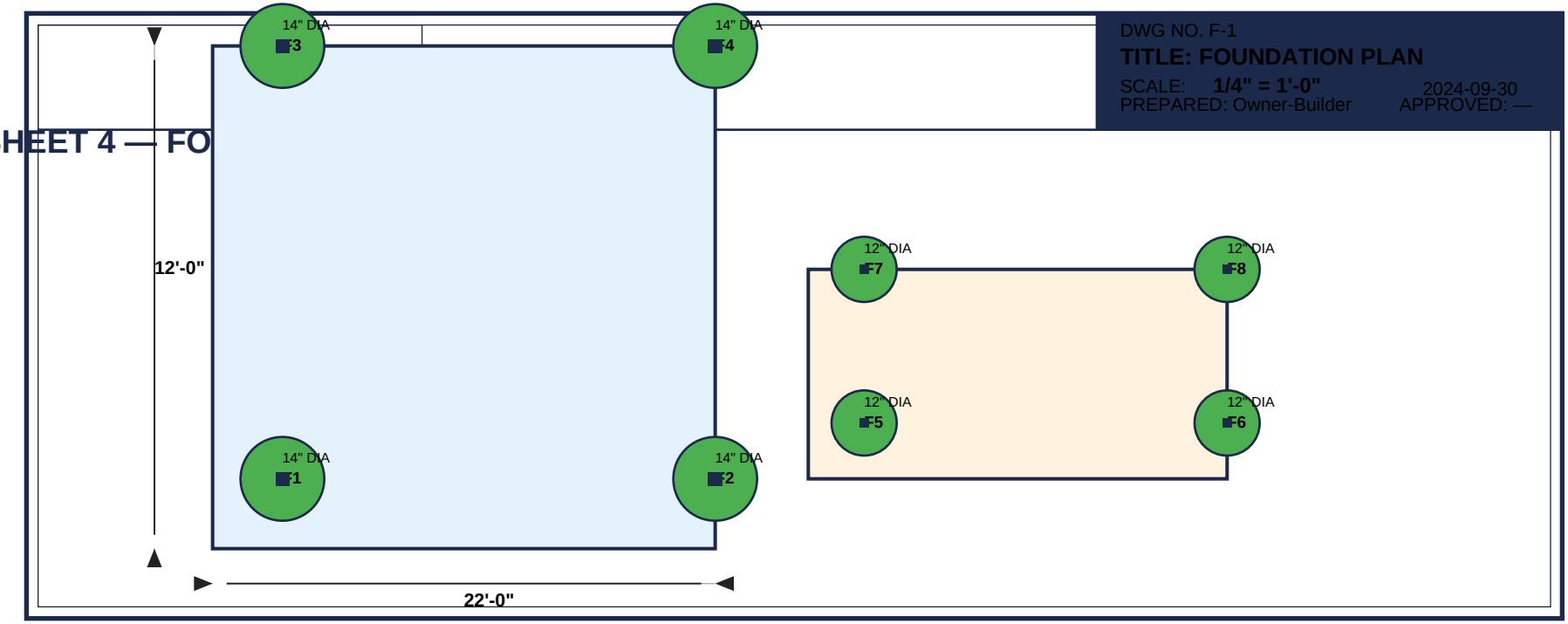
SYMBOL	MEANING	NOTE
1/4" = 1'-0"	Scale notation	Site plans, elevations
1/2" = 1'-0"	Scale notation	Plan views
3/4" = 1'-0"	Scale notation	Connection details
1" = 1'-0"	Scale notation	Material details

SYMBOL	MEANING	NOTE
1-1/4" STEEL PIPE	Nominal size	ASTM A500 Grade B
6x6 PT SPS #2	Lumber size and grade	Southern Pine Structural #2
4x6 PT SPS #2	Lumber size and grade	Southern Pine Structural #2
3000 PSI CONC.	Concrete strength	5% air entrainment
SIMPSON GB663	Post base connector	Per Simpson spec
1/2"x8" HDG LB	Lag bolt	Hot-dip galvanized
U-BOLT CLAMP	Pipe clamp	1/2" hex nuts, HDG
Footing @ 48" D	Foundation depth	Min. frost depth



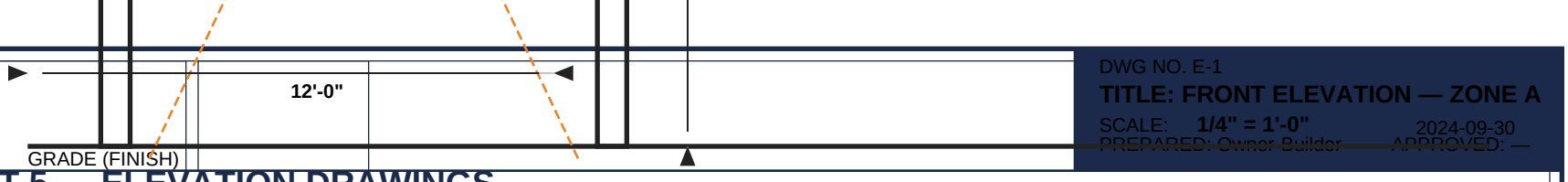
SETBACK COMPLIANCE: All structures maintain minimum 5 ft rear/side yard setback per Bridgewater Zoning Ordinance Section 4.4.1. Total footprint of 264 sq ft is under 300 sq ft exemption threshold (no special permit required). Maximum height of 14 ft is within 25 ft accessory structure limit (Sec. 4.2.3).

SHEET 4 — FO



FOOTING ID	ZONE	POST SIZE	HOLE DIAM	DEPTH	CONCRETE	LOAD CAP.
F1 — F4	A	6x6	14" dia	48" min.	3000 psi	~8,000 lb/ft
F5 — F8	B	4x4	12" dia	48" min.	3000 psi	~5,000 lb/ft

NOTES: All footings to extend minimum 48" below grade (per 780 CMR R403.1.1 frost depth). 2" gravel base at footing bottom. 3000 psi concrete, 5% air entrainment (per IRC R403.1). Grade sloped away from structure minimum 2% (1/4" per ft).



DWG NO. E-1

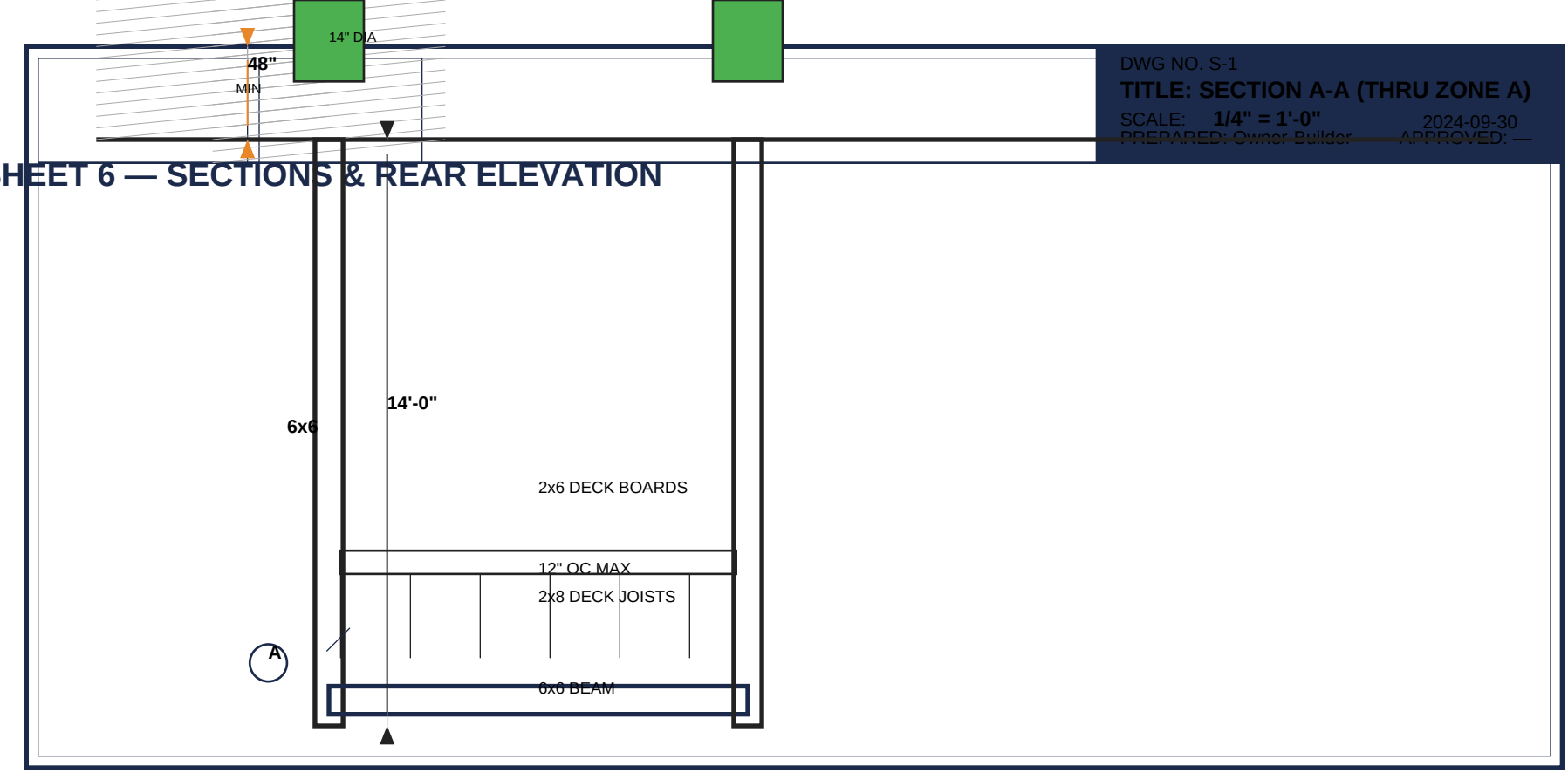
TITLE: FRONT ELEVATION — ZONE A

SCALE: **1/4" = 1'-0"** 2024-09-30

PREPARED: Owner Builder APPROVED: —

SHEET 5 — ELEVATION DRAWINGS

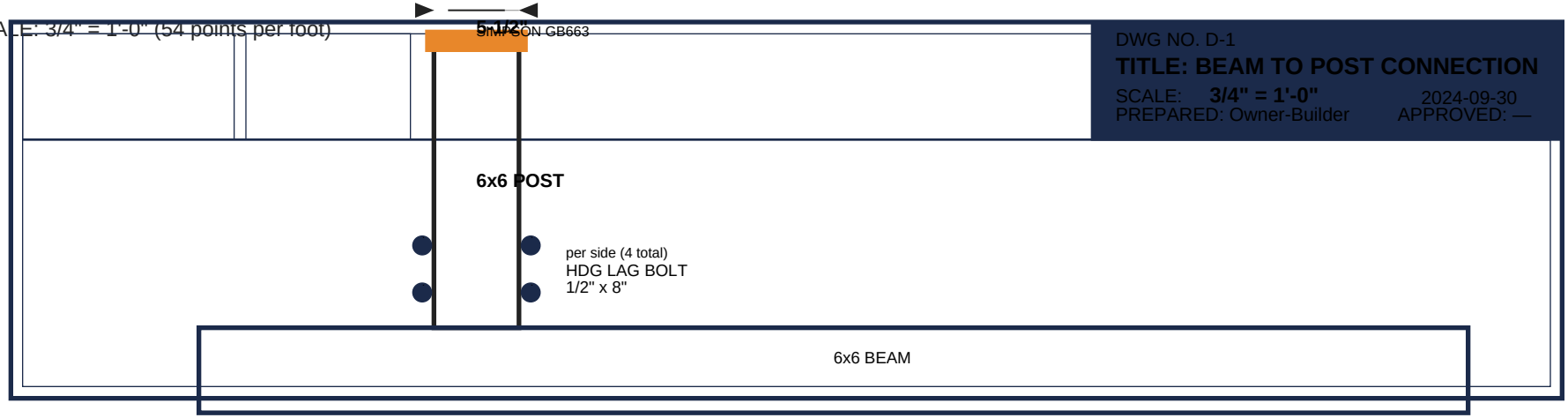
SHEET 6 — SECTIONS & REAR ELEVATION



MEMBER	SIZE	LENGTH	MATERIAL	NOTES
Posts (F1-F4)	6x6	14 ft	SP #2 GC PT	Frost depth 48"
Posts (F5-F8)	4x4	12 ft	SP #2 GC PT	Frost depth 48"
Beams (Zone A)	6x6	12 ft	SP #2 GC PT	Main load-bearing
Beams (Zone B)	4x6	10 ft	SP #2 GC PT	Mon/ bar support
Joists	2x8	6 ft	SP #2 GC PT	12" OC max
Deck boards	2x6	12 ft	SP #2 GC PT	1/4" gap b/w boards

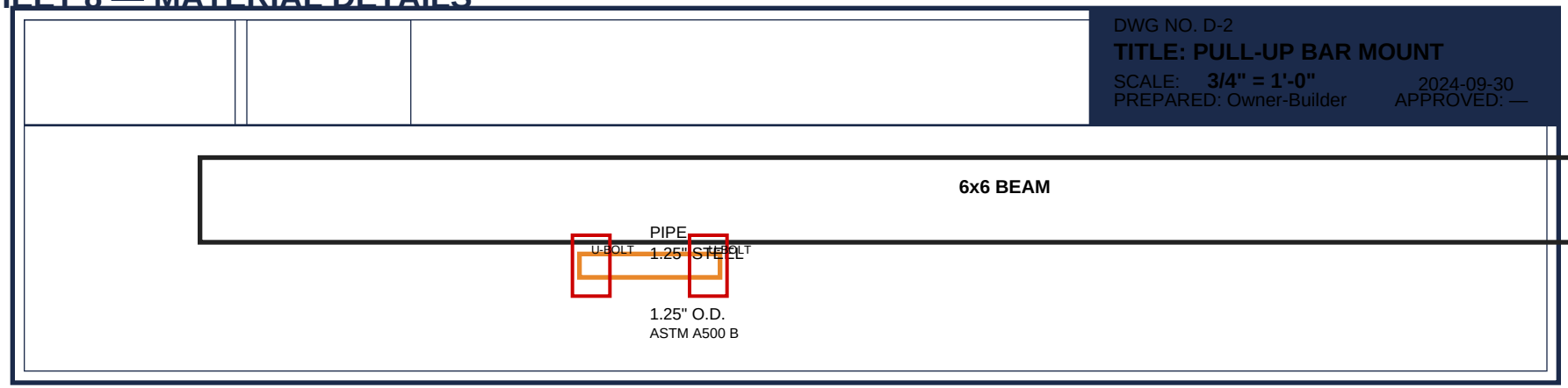
SHEET 7 — CONNECTION DETAILS

SCALE: 3/4" = 1'-0" (54 points per foot)



FASTENERS: 2 x 1/2" x 8" HDG lag bolts per side (4 total per post) + 1 Simpson GB663/664 post base connector per post. All bolts per Simpson Strong-Tie specifications.

SHEET 8 — MATERIAL DETAILS



FASTENERS: 2 U-bolt clamps per end (4 total). Each clamp: 1/2" diameter, 1.25" bore, 1/2" hex nuts + flat washers. Pipe: 1-1/4" ASTM A500 Grade B, 46 ksi yield.

SHEET 9 — BILL OF MATERIALS

ITEM	QTY	SIZE	GRADE	EST. UNIT	EST. TOTAL
6x6 Posts (14 ft)	4	14 ft	#2 GC PT	$85	$340
6x6 Posts (12 ft)	4	12 ft	#2 GC PT	$70	$280
6x6 Beams	8	12 ft	#2 GC PT	$70	$560
4x6 Beams	4	12 ft	#2 GC PT	$45	$180
4x4 Posts	6	8-12 ft	#2 GC PT	$25	$150
2x6 Joists	10	12 ft	#2 GC PT	$18	$180
2x6 Deck boards	4	12 ft	#2 GC PT	$18	$72
2x4 Braces	4	12 ft	#2 GC PT	$12	$48

LUMBER SUBTOTAL: ~$1,810

ITEM	QTY	SPEC	TYPE	EST. UNIT	EST. TOTAL
1/2"x8" Lag Bolts	48	HDG	Lag bolt	$1.20	$58
1/2" Hex Nuts	48	HDG	Nut	$0.30	$14
3/8" Hex Bolts	24	HDG G4	Hex bolt	$1.50	$36
3/8" Hex Nuts	24	HDG	Nut	$0.30	$7
3" Deck Screws	400	#8 Coated	Screw	$0.08	$32
Simpson GB663/664	4	—	Post cap	$12	$48
Simpson HD8	8	—	Hurricane tie	$4	$32
Simpson JH	20	—	Joist hanger	$3	$60
U-Bolt Clamps	24	1.25" HDG	Clamp	$2.50	$60

HARDWARE SUBTOTAL: ~$347

ITEM	QTY	SIZE	TYPE	EST. UNIT	EST. TOTAL
Quikrete Fast-Set	40	80 lb	—	$6.50	$260
Gravel bags	8	50 lb	Drainage	$5.00	$40
Rubber Mulch	15	2 cu ft	Playground	$12	$180
Rubber Mats	4	2x2 ft	Interlocking	$25	$100

CONCRETE & GROUND SUBTOTAL: ~$580

ITEM	QTY	SIZE	TYPE	EST. UNIT	EST. TOTAL
Steel Pipe (pull-up)	1	1.25"x8 ft	A500 B	$45	$45
Steel Pipe (dip)	2	1"x3 ft	A500 B	$25	$50
Safety Netting	2	10x200 ft	UV-stabilized	$40	$80
UV Zip Ties	500	3"	UV-rated	$0.03	$15
Shade Sails	2	14x14x14 ft	HDPE UV	$120	$240
Turnbuckles	4	6" min	Stainless	$8	$32
Eye Bolts	4	5/8"x6"	Stainless	$6	$24
Fibershed Rope	200 ft	3/4"	Climbing	$0.40/ft	$80
Rope Discs	6	—	Hanging obs.	$15	$90
Rope Clips	20	—	Stainless	$1.50	$30
Carabiners	10	—	12k+ lb	$10	$100
Swing Chains	2 sets	7 ft	S-hook	$20	$40
Bungee Cord	1 roll	3/4"x25 ft	—	$18	$18
Steel Primer	1	—	Red oxide	$8	$8
Deck Stain	3 gal	—	Exterior oil	$30	$90

ITEM	QTY	SIZE	TYPE	EST. UNIT	EST. TOTAL
Copper Naphthenate	1 qt	—	Preservative	$12	$12
Wood Filler	4 tubes	—	Exterior	$3	$12

STEEL, ROPE & RIGGING SUBTOTAL: ~$981
LUMBER: ~$1,810 | **HARDWARE:** ~$347 | **CONCRETE:** ~$580 | **STEEL/ROPE:** ~$981
TOTAL ESTIMATED COST: ~$3,718 (range $3,500—$3,900)

BUDGET OPTIMIZATION:
– Buy lumber in bulk from wholesale — saves ~15-20%
– Purchase Simpson connectors in bulk packs — saves ~10%
– Shop for shade sails at discount (Amazon, Wayfair, clearance)
– Use recycled rubber mulch from local playground suppliers
– Skip rope discs if budget is tight — saves ~$90

SHEET 10 — CONSTRUCTION CHECKLIST

PRE-CONSTRUCTION

– Call 811 (Mass. utility locators) — minimum 3 business days
– Obtain zoning determination from Bridgewater Building Dept
– Complete Homeowner Waiver Form (owner-builder)
– Submit online permit application with full drawing set

DO NOT START WORK WITHOUT RECEIVING BUILDING PERMIT CARD

– Mark post locations with spray paint
– Verify property lines and setbacks
– Order all materials per BOM
– Apply copper naphthenate to all cut ends before construction

FOUNDATION

– Dig all 8 footings to 48" depth minimum
– Verify footing spacing: Zone A 12x12, Zone B 10x5
– Add 2" gravel base to each hole
– Schedule footing inspection
– Pour concrete — 3000 psi, 5% air entrainment
– Cure minimum 72 hours before framing
– As-built photos to building@bridgewaterma.org

FRAMING

– Set posts plumb within 1/4" in 10 ft
– Verify all posts square (diagonals equal)
– Install post connectors (Simpson GB663 / HD8)
– Install beams — level within 1/4" in 10 ft
– Install cross-braces at 45 degrees
– Install upper deck joists at 12" OC (2x8 at 16" OC max)
– Install deck boards — 1/4" gap between boards
– Schedule framing inspection
– Verify all connections are tight

EQUIPMENT & FINISH

– Install pull-up bar — test with 200 lb dynamic load

– Install monkey bar rungs — test each with 150 lb
– Install dip station — verify both heights level
– Install rope tower — secure top and bottom
– Install swing chains — verify S-hook security
– Install safety netting — check all ties tight
– Install shade sails — tension with turnbuckles
– Install climbing holds — verify lag bolt security
– Apply stain/sealant — minimum two coats
– Install ground surface — rubber mulch/tiles
– Apply copper naphthenate to all field-cut ends

FINAL INSPECTION

– Call for final inspection
– Verify all bolts torqued and secured
– Verify netting gaps less than 2 inches
– Verify overhead clearance greater than 15 ft

SHEET 11 — BRIDGEWATER PERMIT REQUIREMENTS

Permit Type: Building Permit — Accessory Structure

REQUIRED SUBMITTAL DOCUMENTS

– Online building permit application (Bridgewater portal)
– Plot plan showing proposed structure with setbacks
– Foundation details (this drawing set)
– Framing details with cross sections (this drawing set)
– Elevations (this drawing set)
– Homeowner Waiver Form (owner-builder)
– Contractor license (if contractor applies)

FEE SCHEDULE (Bridgewater, effective 7/1/2024)

150-300 sq ft: $12 per $1,000 of valuation — Minimum $200
Estimated valuation: ~$4,000
Estimated permit fee: ~$48-50

INSPECTIONS REQUIRED

– Footing/Foundation — before concrete is poured
– Framing — after structural assembly complete
– Final — after all work complete including finishes

TIMELINE

– Application processing: 5-10 business days
– Permit card issued before work begins
– Inspections must be scheduled in advance
– Email photos to building@bridgewaterma.org

IMPORTANT NOTICE

This drawing set is provided as a guide for owner-builder construction. All work must comply with the latest adopted codes and local ordinances of the Town of Bridgewater, MA. For structures exceeding 12 ft in height or complex load paths, consultation with a Massachusetts-licensed structural engineer is recommended. No warranty is made regarding the accuracy or completeness of this document. The owner-builder assumes all responsibility for compliance with applicable codes, regulations, and safety standards.

END OF STRUCTURAL DRAWING SET

Prepared using IBC 2018/IRC 2018 design criteria, 780 CMR 10th Edition, and Bridgewater zoning ordinance Section 4.4.